

Lentiviral Production

Introduction

Based on Sofia's existing protocols: [Lentivirus production](#) and [Lipofectamine3000](#)

Refer to the table [here](#) to if changing container

Can use in combination with plasmid packaging excel but there is a provided calculation table below

Materials

- > Opti-MEM reduced serum medium
- > HEK293FT cell culture
- > psPAX2 plasmid (Addgene 11260)
- > pDM2.G plasmid (Addgene 12259)
- > Transfer plasmid (your plasmid)
- > Lipofectamine reagents
 - > L3000
 - > P3000
- > PBS
- > DMEM medium
- > Millipore low protein binding sterile filter 0.45um
 - > in bacterial room storage room shelves
- > Lenti-X concentrator

Procedure

Day -3 to -1: Split cells and prepare plasmid mix

1. Split cells (usually HEK293FT) and seed in T75 flask with DMEM media
 - Want them to be 80% confluency for transfection the next day
 - If splitting on Friday to tranfect on Monday, split 1:15
2. Prepare plasmid mix according to Lenti-packaging excel (current experiment sheet)
 - Contains transfer plasmid, psPAX2 and pMD2.G for a total of 25 ug

Plasmid mix calculator							
	A	B	C	D	E	F	G
1	Plasmid	Size (bp) <-- enter	MW (g/mol) = bp × 660	Molar fraction (1/sum of MW) (%)	µg needed (fraction × total)	Stock conc (ng/µL) ← enter	Volume to add (µL)
2	psPAX2	10712	7069920	0.43	10.75	275.0	39.1
3	pMD2.G	5822	3842520	0.234	5.85	469.0	12.5
4	Transfer vector	8364	5520240	0.336	8.4	165.0	50.9
5	Total	—	16432680	1	25		102.5

Day 0: Lipofectamine 3000 transfection

3. Bring Opti-MEM to room temperature

4. Check cell confluency

5. Prepare tube A and tube B (below amounts are for T75)

Tube A: 750 µL Opti-MEM + 59.2 µL Lipofectamine3000

Tube B: 750 µL Opti-MEM + 40 µL P3000 + 25 µg plasmid mix

Notes: L3000 is 1:1 with DNA and P3000 is 2:1 with DNA

then why these amounts, ask sofia?

6. Mix both tubes separately by inversion and add tube A to tube B, mix by inversion

7. Incubate for 15 minutes at RT

8. Remove media from cells (in T75) and wash with PBS gently

To remove media, tilt flask with cap up to pile media on the corner opposite the adherent cells

Add PBS in the same fashion, move PBS around flask to wash and then remove in from the same corner

9. Add appropriate amount of Opti-MEM media to flask (12 mL in this case)

verify with sofia

10. Add all of the transfection mix, drop-wise to the cells without detaching them

Try to spread it around the whole flask as you add

11. Remove and replace media after 6 hours

Check name of media but it is specific

Day 2: First harvest

12. Collect supernatant (contains virus) by tilting as done before and store in 50 mL falcon tube in dridge

13. Replace with 12 mL DMEM media

Day 3: Second harvest and concentration

14. Collect virus containing supernatant and pool with the first harvest

15. Centrifuge at 500g for 10 min at 4 °C to pellet cell debris

Set centrifuge back to RT after

16. Filter virus particles through a millipore low protein binding sterile filter 0.45 µm

Virus size range between 60-100 nm in diameter usually

DO NOT use nitrocellulose filter

17. Add 1:3 volume of Lenti-X concentrator = 24 ml supernatant + 8 ml Lenti-X

Mix by inversion and store at 4 °C overnight

Day 4: Aliquot and store virus

18. Centrifuge virus at 3000xg for 60 min at 4°C

off-white pellet will form

19. Carefully remove the supernatant, taking care not to disturb the pellet

Pour out into waste container and tap on paper towel lightly

20. Resuspend in 1:100 DMEM media to original volume (24 mL) = 240 µl

100X concentration

21. Make aliquots of around 50 µl and place in -80°C

Usually distribute evenly between 4 tubes